1) **Show that**

a) $\delta\mu = \frac{1}{2}(\Delta + \boldsymbol{\nabla})$

We need to show the given identity using finite difference operators.

- Forward Difference (Δ):

  $\Delta \boldsymbol{f}(x) = \boldsymbol{f}(x + \hbar) - \boldsymbol{f}(x)$

- Backward Difference (∇):

  $\nabla \boldsymbol{f}(x) = \boldsymbol{f}(x) - \boldsymbol{f}(x - \hbar)$

- Central Difference (δ):

  $\delta \boldsymbol{f}(x) = \frac{\boldsymbol{f}(x+\hbar) - \boldsymbol{f}(x-\hbar)}{2}$

- **Mean Operator (μ)**:

  $\mu \boldsymbol{f}(x) = \frac{\boldsymbol{f}(x+\hbar) + \boldsymbol{f}(x-\hbar)}{2}$

Now, rewriting $\delta \boldsymbol{f}(x)$:

$\delta \boldsymbol{f}(x) = \frac{\boldsymbol{f}(x+\hbar) - \boldsymbol{f}(x-\hbar)}{2} = \frac{1}{2}(\Delta f(x) + \boldsymbol{\nabla} f(x))$

Thus, $\delta\mu = \frac{1}{2}(\Delta + \nabla)$

b) Δ **-** ∇ **=** $\Delta\nabla$

Expanding the finite difference operators:

$(\Delta - \nabla)f(x) = [f(x + h - f(x)] - [f(x) - f(x - h)] = f(x + h) - 2f(x) + f(x - h)$

We recognize this as the product of forward and backward differences:

$\Delta\nabla f(x) = (f(x + h) - f(x))(f(x) - f(x - h))$

Thus, $\Delta - \nabla = \Delta\nabla$

2. Find Lagrange's interpolation polynomial fitting the points and find $y(5)$

Given points:

$(1, -3), (3, 0), (4, 30), (6, 132)$

Using the Lagrange Interpolation Formula,

$$P(x) = \sum_{i=0}^{n} y_i L_i(x)$$

$$L_i(x) = \prod_{j=0, j\neq 1}^{n} \frac{x - x_j}{x_i - x_j}$$

Computing the Lagrange basis polynomials:

$$L_1(x) = \frac{(x-3)(x-4)(x-6)}{(1-3)(1-4)(1-6)}$$

$$L_2(x) = \frac{(x-1)(x-4)(x-6)}{(3-1)(3-4)(3-6)}$$

$$L_3(x) = \frac{(x-1)(x-3)(x-6)}{(4-1)(4-3)(4-6)}$$

$$L_4(x) = \frac{(x-1)(x-3)(x-4)}{(6-1)(6-3)(6-4)}$$

Plugging in the given values:

$P(x) = (-3)L_1(x) + (0)L_2(x) + (30)L_3(x) + (132)L_4(x)$

Now, substituting $x = 5$ and solving, we get:

$y(5) \approx 62$

Final answer: $y(5) = 62$

3. Evaluate $f(15)$ using the given table of values.

Given values:

x	10	20	30	40	50
$y = f(x)$	46	66	81	93	−

Using linear interpolation between $(10, 46)$ and $(20, 66)$:

Using the formula:

$$f(x) = f(a) + \frac{x-a}{b-a} \times (f(b) - f(a))$$

Substituting:

$$f(15) = 46 + \frac{15-10}{20-10} \times (66 - 46)$$

$$f(15) = 46 + \frac{5}{10} \times 20$$

$$f(15) = 46 + 10 = 56$$

Final answer: $f(15) = 56$

4) Find the equation of the best-fitting straight line for the given data.

Given data:

$$X = [1, 3, 4, 6, 8, 9, 11, 14]$$

$$Y = [1, 2, 4, 4, 5, 7, 8, 9]$$

The equation of the best-fitting straight line is:

$$Y = a + bX$$

Where a $(intercept)$ and b $(slope)$ are calculated using:

$$b = \frac{n\sum XY - \sum X \sum Y}{n\sum X^2 - (\sum X^2)}$$

$$a = \frac{\sum Y - b\sum X}{n}$$

Let’s compute the required values:

$$\sum X = 1 + 3 + 4 + 6 + 8 + 9 + 11 + 14 = 56$$

$$\sum Y = 1 + 2 + 4 + 4 + 5 + 7 + 8 + 9 = 40$$

$$\sum XY = 1 + (3 \times 2) + (4 \times 4) + (6 \times 4) + (8 \times 5) + (9 \times 7) + (11 \times 8) + (14 \times 9)$$

$\sum XY$ = 364

$$\sum X^2 = 1^2 + 3^2 + 4^2 + 6^2 + 8^2 + 9^2 + 11^2 + 14^2 = 524$$

By keeping these values in the above equations we get:

$b = 0.636,\ a = 0.548$

Final answer: $Y = 0.548 + 0.636X$

5) For what values of λ and μ, the system of equations may have different types of solutions?

Given system of equations:

$x + y + z = 6$

$x + 2y + 3z = 10$

$x + 2y + \lambda z = \mu$

To determine when this system has a unique solution, infinite solutions, or no solution, we analyze the coefficient matrix:

$A =$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix}$$

The determinant of A determines whether the system is consistent or inconsistent.

$det(A) =$

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix}$$

Expanding along the first row:

$$det(A) = 1 \times \begin{vmatrix} 2 & 3 \\ 2 & \lambda \end{vmatrix} - 1 \times \begin{vmatrix} 1 & 3 \\ 1 & \lambda \end{vmatrix} + 1 \times \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = 2\lambda - 6 - \lambda + 3 = \lambda - 3$$

Final Answer:

Unique solution: If $\lambda \neq 3$.

Infinite solutions: If $\lambda = 3$ and $\mu = 10$.

No solution: If $\lambda = 3$ and $\mu \neq 10$.

6) Solve using Euler's method for $x = 0.2$ with step size $h = 0.1$.

Given Differential Equation: $\frac{dy}{dx} = 1 - y$

with the assumed initial condition: $y(0) = 1$

Euler's method formula: $y_{n+1} = y_n + hf(x_n, y_n)$

Where $f(x, y) = 1 - y$.

Compute $y(0.1)$: $y_1 = y_0 + 0.1(1 - y_0) = 1 + 0.1(1 - 1) = 1$.

Compute $y(0.2)$: $y_2 = y_1 + 0.1(1 - y_1) = 1 + 0.1(1 - 1) = 1$.

Final answer: $y(0.2) = 1$.